Accurate Recycled FPGA Detection Using an Exhaustive-Fingerprinting Technique Assisted by WID Process Variation Modeling

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***Abstract*—In this study, a novel method for the detection of recycled field-programmable gate arrays (FPGAs) is proposed. This method is based on with-in die (WID) process variation model over an exhaustive path characterization [referred to as exhaustive-fingerprint (X-FP)]. In the proposed method, X-FP is capable of fully characterizing frequencies on all paths in look-up tables (LUTs) using advanced ring oscillator (RO) design to exhaustively capture deterioration by aging. Although machine learning (ML)-based classification is often used for recycled FPGA detection, X-FP yields a large amount of measurement data, which cannot be appropriately handled by typical ML algorithms, if they are used as a feature vector. The proposed method utilizes the model parameters extracted by WID variation modeling as the feature vector in the ML algorithm. These model parameters simply and accurately represent the process variation for each FPGA. In this way, the ML-based fresh/recycled classification works very well with the simple feature vector. Experiments using 50 commercially available FPGAs reveal that X-FP can capture the degradation effects, which cannot be detected by conventional methods. Moreover, the WID modeling achieves 99.6% feature size reduction per one FPGA. It also demonstrates that the model parameters exhibit good distance properties between fresh and aged FPGAs. Additionally, the ML-based classification which uses a one-class support vector machine can successfully detect 2 aged FPGAs (48 h accelerated aging) without any misclassification and another 4 aged FPGAs (24 h accelerated aging) with a very few misclassifications as fresh FPGAs.**

***Index Terms*—Aging degradation, feature engineering, feature vector, field-programmable gate array (FPGA), process variation, recycling, with-in die (WID) modeling.**

I. Introduction

THE ECONOMIC globalization and the unprecedented expansion of the integrated circuits (ICs) supply chain have resulted in the usage of recycled ICs, which is a rising concern. This scenario not only has an impact on the business of the IC industry but also raises concerns in some critical applications, such as automobile, medical, and communication systems due to the reduced reliability of the recycled ICs [1]. Recyclers often camouflage previously used ICs as fresh and shift them to the IC supply chain. The leakage of these recycled ICs makes them more vulnerable to failure in some critical applications [2]. Either their usage is intentional or unintentional. Meanwhile, field-programmable gate arrays (FPGAs) have been widely used in modern electronic systems due to their low development cost and short time-to-market. Additionally, their recent usage as accelerators for artificial neural networks has become a new trend [3], [4]. Thus, recycling of FPGAs is a major concern taking into account their increasing usage [5].

Even though recycled FPGAs exhibit good functionality initially, their performance decrease quickly due to the aging effect arising from their previous usage. However, it is difficult to stop the penetration of recycled FPGAs into the IC market due to the large scale expansion of complex IC supply chain. In the literature, only a small number of studies to date have concentrated on the identification of recycled FPGAs. Most of the recycled IC detection methods have been focused on application-specific ICs (ASICs) [6]–[10]. All of these studies are sensor-based methods, which result in hardware overhead costs and rely on a statistical model to detect recycled ICs.

Certain methods, which take into account the increase in path delay induced by aging, have been proposed for the purpose of detection of recycled FPGAs [5], [11], [12]. In [5], the basic idea and key concept of detection of recycled FPGAs using ring oscillator (RO) frequencies were presented. Since the performance of the configurable logic blocks (CLBs) in an FPGA degrades with usage, the frequency of the RO also decreases [13]. Using a one-class support vector machine (OC-SVM) [14], the performance degradation between recycled and fresh FPGAs can be effectively detected. In [11], path delays in a look-up table (LUT) are exhaustively characterized using RO frequencies. While the method proposed in [11] uses only a certain number of CLBs in an FPGA for detection, the method proposed in [12] utilizes the frequencies on all CLBs in the FPGAs, thus producing a "fingerprint" for each FPGA. These frequencies are stored into a database, which is used for fresh FPGAs before shipping. Although the process variation and spatial correlation can be effectively captured in [12], compared with the method proposed in [11] the method does not consider all the paths in the LUT.

Manuscript received March 24, 2020; revised June 7, 2020 and July 30, 2020; accepted September 6, 2020. Date of publication September 14, 2020; date of current version July 19, 2021. This work was supported in part by the Telecommunications Advancement Foundation and in part by Japan Society for the Promotion of Science KAKENHI Grant 18K18025. This article was recommended by Associate Editor W. Zhang. *(Corresponding author: Foisal Ahmed.)*

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Digital Object Identifier 10.1109/TCAD.2020.3023684

In this article, an exhaustive fingerprint analysis method for the accurate detection of recycled FPGAs is proposed. In this method, the frequencies for all paths of LUTs in all CLBs are taken into account. Hereafter, we refer to it as exhaustive-FP, or exhaustive-fingerprint (X-FP) in short. X-FP is based on the combination of the two techniques described in [11] and [12]. That is, X-FP yields differential fingerprints for each path. In this way, the aging-induced degradation can be accurately detected, and thus improved detectability of the recycled FPGAs can be achieved. However, avoidably, X-FP drastically leads to a large number of measured frequencies, which is the dimensionality of a feature vector in machine learning (ML) recognition. This, in turn, results in the severe degradation of the accuracy in the ML-based detection due to "curse of dimensionality" problem. Moreover, the establishment of the fingerprint database becomes unrealistic in terms of database maintenance cost.

To deal with the above problem, we extend our previous work [15] on the feature engineering for the recycled FPGA detection, which is based on a with-in die (WID) modeling, to effectively apply proposed X-FP. Feature engineering is a process to generate or extract new features from raw data so as to improve the performance of ML algorithm, and it can make ML model more flexibility, faster in detection accuracy calculation, and to avoid the difficulties of overfitting or curse of dimensionality problem as well. In particular, feature engineering based on a specific domain knowledge contributes extensively to performance improvement [16]. The proposed feature engineering exploits an established technique of large-scale integration (LSI) process variation modeling.

In the proposed method, WID modeling is conducted for each fingerprint obtained by X-FP. WID process parameter variations in manufactured LSIs are generally divided into "random" and "systematic" components [17]. The random component originates from random phenomena, such as random dopant fluctuation (RDF) and line edge roughness (LER) [18]. In contrast, the systematic component represents a deterministic trend on a die and is often modeled by a polynomial of geometrical locations on the die [19], [20]. However, the determination of its maximum order is not trivial in the WID modeling. The proposed method determines it on the basis of Akaike's information criterion (AIC), as proposed in [21]. In the proposed recycled FPGA detection, the model parameters extracted from the WID modeling (e.g., the coefficients of the polynomial) are used as the feature vector in the ML algorithm. Since the model parameters summarize well the process variation correlation on the die, the effective recycled FPGA detection becomes possible. Although principal component analysis is generally used [5], [22] for the purpose of the dimensionality reduction, our feature vector engineering is more fundamental and objective because it is based on the well-studied process variation modeling of manufactured LSIs [21], [23], [24].

While the feature engineering through the WID modeling is based on our previous work [15], here the conventional recycled FPGA detection methods are discussed in far more detail and the X-FP analysis is newly proposed in order to improve the detectability of recycled FPGAs. The major contributions of this research work are highlighted as follows.

1) The proposed recycled FPGA detection method employs X-FP, which allows the examination of the aging-induced delay characteristics of all paths in all LUTs, to achieve accurate detection.
2) The WID modeling-based feature engineering is capable of reducing not only the size of the feature vectors but also the database maintenance cost. The extracted model parameters are effectively used in the ML-based fresh/recycled classification.
3) Experiments on 50 commercial FPGAs (including aging acceleration to 2 FPGAs) demonstrate various path differences between fresh and aged FPGAs by the X-FP characterization.
4) These experiments also demonstrate that model parameters vary in the aging acceleration and aged FPGAs are successfully detected using OC-SVM. In contrast, the conventional method leads to misclassification.

The remainder of this article is organized as follows. The related research work is briefly presented and the motivation for the proposed work is discussed in Section II. The proposed recycled FPGA detection method, including X-FP analysis and WID-modeling-based recycled FPGA detection method, is described in Section III. The experimental procedure and results are presented in Section IV. Finally, the conclusions from this article are presented in Section V.

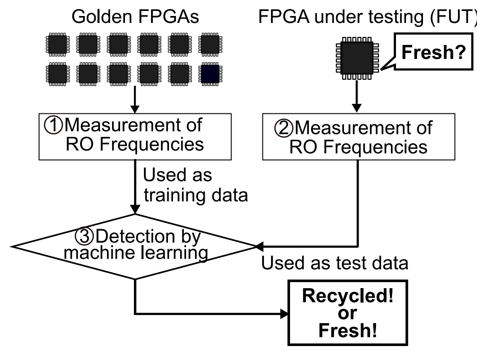


Fig. 1. Flowchart of the recycled FPGAs detection method proposed in [5].

II. Related Research Work and Findings

Initially, three previous studies on recycled FPGAs detection [5], [11], [12] are briefly summarized. The basic idea of the recycled FPGAs detection has been described in [5] and a flowchart of this method is shown in Fig. 1. In [5], the existence of golden (i.e., known fresh FPGAs) is assumed. If an FPGA under testing (FUT) has been previously used, some of its RO frequencies will be degraded compared to the RO frequencies of the golden FPGAs. This is due to aging mechanisms, including bias temperature instability (BTI), time dependent dielectric breakdown (TDDB), and hot carrier injection (HCI) [25]–[27]. Comparing the measured frequencies between golden FPGAs and an FUT using an ML algorithm, the FUT can be classified as either recycled or fresh. The flowchart of this method involves the following three parts.

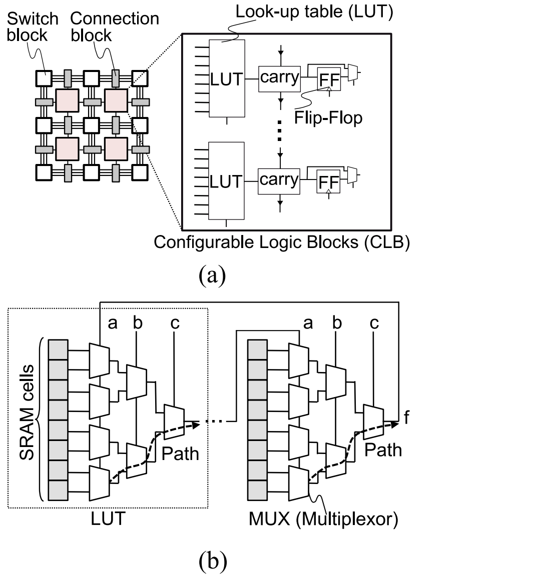


Fig. 2. Basic idea of implementation of RO using LUTs in the FPGA. (a) Typical structure of an FPGA. (b) Logic structure of LUT.

1) Inverter-based ROs are designed on all golden FPGAs and those frequencies are measured to form a feature vector. Then, a database of frequencies is constructed. The ML model learns a feature of the golden FPGAs using the database as training data.
2) Before using an FUT in a system, its frequencies are measured according to part 1).
3) The trained model tests whether the FUT is fresh or not, using the frequencies measured in part 2).

Other methods [11], [12] also follow this concept.

Before describing the previous studies reported in [11] and [12], the typical structure of a commercial FPGA is briefly discussed using Fig. 2. The CLB, connection block, and switch block are placed to form an array in an FPGA. A CLB consists of several LUTs, carry chain, and flip-flops as shown in Fig. 2(a). Various logic functions can be created in a CLB by changing the values in the SRAM cells and the path of the LUT. A path of the LUT is defined as the route from the nearest multiplexer (MUX) to the SRAM cells at the output of the LUT. The CLBs are connected to the switch block through the connection blocks. Multistage RO can be designed in a single CLB using multiple LUTs, as shown in Fig. 2(b). The frequency of an RO can be measured by a counter placed at the feedback path.

The distinct difference among the three methods described in [5], [11], and [12] is RO characterizations. The method described in [11] exhaustively characterizes all paths in some CLBs in order to characterize all the delay variations on all paths. This is called exhaustive path characterization and it is called XP hereafter. Using XOR and XNOR gates and specifying the inputs of LUTs appropriately, all paths inside the LUTs can be accessed and the value of a selected SRAM cell will propagate. In the RO, the logic gates XOR and XNOR operate as an inverter. For an i-inputs LUT, 2^{i-1} paths can be analyzed since one of the inputs is required to form the inverter chain of the RO.

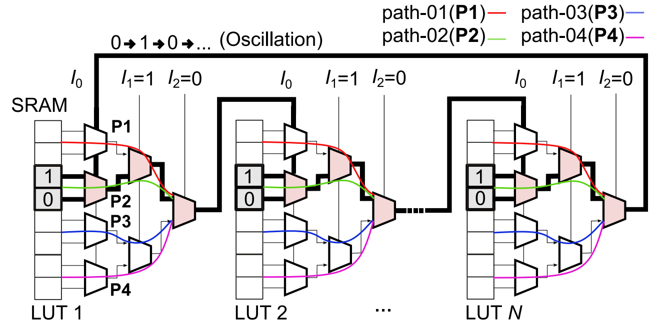


Fig. 3. N-stage XOR-based RO covers paths-02 (highlighted lines) by setting $I_1 = 1$ and $I_2 = 0$.

TABLE I
CONFIGURATION OF THE EXHAUSTIVE PATHS USING THE XNOR AND XOR LOGIC FUNCTIONS. I_1 AND I_2 ARE SET TO APPROPRIATE VALUES

Configuration I_1 I_2	Configuration Function	Activation
00	XNOR	path-01
11	XNOR	path-04
10	XOR	path-02
01	XOR	path-03

The configuration of an RO circuit with N (odd-number) stages is shown in Fig. 3. Each LUT has three input pins I_0, I_1, and I_2 in this example. The output of an LUT is connected to the input pin I_0 of the next LUT and a closed feedback loop is formed to create the RO. The input pin I_0 and the related signal will oscillate in this example. All the configurations required to activate all the paths using the XOR- and XNOR-based ROs are presented in Table I. In this table, 4 ($= 2^{3-1}$) paths (path-01 to path-04) are composed in total, since $i = 3$. For instance, an XOR-based RO can be created to cover path-02 by setting I_1 and I_2 to "1" and "0," respectively. The values of SRAM cells, "1" and "0" propagate alternately to the next LUT stage. Then, an oscillation is generated as highlighted in Fig. 3.

Let f be the measured frequency of the corresponding RO. The frequencies for a single CLB, $\boldsymbol{F}$, can be represented as $[f_1, f_2, \ldots, f_{2^{i-1}}]$. In the XP characterization, the target CLBs are given in advance by the manufacturer. A frequency array $\boldsymbol{F}_{\text{XP}}$ can be expressed when M CLBs of a single FPGA are analyzed as follows:

$$\boldsymbol{F}_{\text{XP}} = [\boldsymbol{F}_1, \boldsymbol{F}_2, \ldots, \boldsymbol{F}_M]. \tag{1}$$

An ML algorithm is used to learn from this frequency array in order to classify FUTs.

The XP characterization procedure is summarized in Algorithm 1. The input of the algorithm is the M CLB locations and the number of inputs, i, in the CLB. The output is the measured frequency array $\boldsymbol{F}_{\text{XP}}$. As listed in lines 2–5, the frequencies of the ROs in the CLBs are exhaustively measured using XOR- and XNOR-based ROs at the CLB location m. Consequently, this method can fully cover all the paths in a

Algorithm 1 XP Characterization [11]

Require: Locations of the M target CLBs and number of inputs of the LUT i

for $m = 1$ to M **do**
  for $n = 1$ to 2^{i-1} **do**
    Measure f_n of XOR- and XNOR-based RO in the CLB at m
    Insert f_n to $\boldsymbol{F}_m$
  end for ▷ $\boldsymbol{F}_m = [f_1, f_2, \cdots f_{2^{i-1}}]$
  Insert $\boldsymbol{F}_m$ to $\boldsymbol{F}_{\text{XP}}$
end for ▷ $\boldsymbol{F}_{\text{XP}} = [\boldsymbol{F}_1, \boldsymbol{F}_2, \cdots, \boldsymbol{F}_M]$
Return $\boldsymbol{F}_{\text{XP}}$

Algorithm 2 FP Characterization [12]

Require: Total grid information (X, Y)

for $x = 1$ to X **do**
  for $y = 1$ to Y **do**
    Measure $f_{(x,y)}$ of inverter-based RO in CLB at (x, y)
    Insert $f_{(x,y)}$ to $\boldsymbol{F}_{\text{FP}}$
  end for
end for ▷ $\boldsymbol{F}_{\text{FP}} = [f_{(1,1)}, f_{(2,1)}, \cdots, f_{(X,Y)}]$
Return $\boldsymbol{F}_{\text{FP}}$

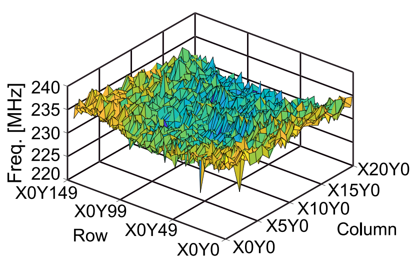


Fig. 4. Example of FPGA fingerprinting (FP) using a commercial FPGA. The frequencies of all CLBs are shown using the grid information in the FPGA [12].

CLB, whereas the tested CLBs need to be carefully selected. The total number of the characterized ROs is $M \times 2^{i-1}$.

Another approach, which uses the frequencies of the LUTs for all locations as a fingerprint for each FPGA has been proposed in [12]. This technique is called FPGA fingerprinting and hereafter it is simply referred to as FP. An example of the fingerprint technique using a commercial FPGA is shown in Fig. 4. In this method, all the CLBs in the FPGA are accessed in order to investigate the spatial correlation of the process variation on a die. As the static process variation is determined by the nature of the manufacturing process [23], a unique pattern behaves as a fingerprint and enables each FPGA to be recognized. Unlike [11], an inverter gate is used to implement the ROs in the FP measurement. Thus, an RO is made of single input inverters and the RO covers only a single path in the LUTs.

The procedure of the FP technique is described in Algorithm 2. Although, in Fig. 4, the CLB information is given according to the Xilinx Artix-7 FPGA layout [28], hereafter, the term X and Y are denoted for labeling the CLB position instead of row and column for simplicity. The input and output of the algorithm are the total geometrical position X and Y of the CLBs in the FPGA and the fingerprinting frequency array $\boldsymbol{F}_{\text{FP}}$, respectively, where $\boldsymbol{F}_{\text{FP}}$ is represented as

$$\boldsymbol{F}_{\text{FP}} = [f_{(1,1)}, f_{(2,1)}, \ldots, f_{(X,Y)}]. \tag{2}$$

As described in the nested two "for" loops of x and y, the RO measurements are conducted for all CLBs. It is noted that, since there are no spaces having no CLBs in a commercial FPGA, the measurements are practically skipped for these spaces. As reported in [11], since paths age differently in the LUTs, an RO circuit implemented using a single path in the LUT may fail to detect an aged path. Hence, a more sophisticated method capable of covering all paths of the LUTs [11] and all LUTs in the FPGA layout [12] is required to effectively detect a recycled FPGA.

III. Proposed Recycled FPGA Detection Method

In order to achieve more accurate recycled FPGA detection, a novel method for detecting the aging effect on all paths in all CLBs is proposed. The proposed method is based on the combination of the two previous methods, XP and FP. Since the CLBs and LUT paths, which were used in a previous usage of an FPGA, are unknown, X-FP, which allows the exhaustive capture of the aging effect, can improve the recycled FPGA detection accuracy. However, this combinational approach leads to an increase in the feature size during the training phase of the ML algorithm. Large size feature vectors often cause deterioration in the ML-based classification due to the curse of dimensionality problem. Furthermore, the establishment of the golden FPGAs database becomes unrealistic in terms of maintenance cost. Thus, to overcome this problem, the proposed method employs feature engineering based on WID modeling, originally proposed in [15], to reduce the number of the feature vectors. It is noted that, the X-FP measurement also leads to an increase in the measurement. However, the cost can be massively reduced by a statistical performance characterization framework, which is called virtual probe technique [29], as proposed in [22].

The flowchart of the proposed recycled FPGA detection method is still based on the conventional method shown in Fig. 1 and is depicted in Fig. 5. The main parts of the proposed flowchart are summarized as follows.

1) Initially, an X-FP measurement is conducted for the golden FPGAs. The measurement yields 2^{i-1} fingerprints per FPGA. Hereafter, the measured frequencies are called exhaustive fingerprints.
2) Then, WID modeling is applied to the exhaustive fingerprints. The model parameters, which are extracted from the WID model, are fed to ML as feature vectors to enable the learning of the golden FPGAs features. The WID modeling is conducted for each fingerprint.
3) A similar X-FP measurement is conducted as described in part 1) for an FUT.
4) Feature engineering through WID modeling is applied to the X-FP of the FUT as described in part 2).

Algorithm 3 X-FP Characterization

Require: Total grid information (X, Y) and number of input of LUT, i

1: **for** $n = 1$ to 2^{i-1} **do**
2:   **for** $x = 1$ to X **do**
3:     **for** $y = 1$ to Y **do**
4:       Measure $f_{n;(x,y)}$ of XOR- or XNOR-based RO in CLB at (x, y)
5:       Insert $f_{n;(x,y)}$ to the array $\boldsymbol{F}_{n;(x,y)}$
6:     **end for**
7:   **end for** $\triangleright\ \boldsymbol{F}_n = [f_{n;(1,1)}, f_{n;(2,1)}, \cdots, f_{n;(X,Y)}]$
8:   Insert $\boldsymbol{F}_n$ to $\boldsymbol{F}_{\text{X-FP}}$
9: **end for** $\triangleright\ \boldsymbol{F}_{\text{X-FP}} = [\boldsymbol{F}_1, \boldsymbol{F}_2, \cdots, \boldsymbol{F}_{2^{i-1}}]$
10: Return $\boldsymbol{F}_{\text{X-FP}}$

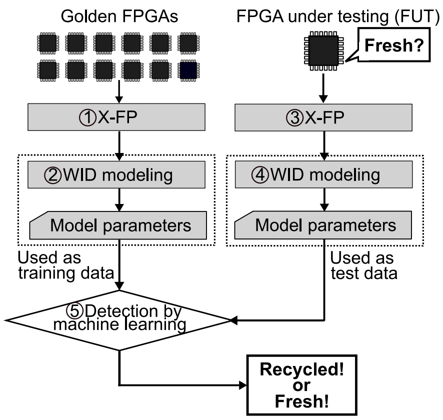


Fig. 5. Overall flowchart of the proposed recycled FPGA detection method.

5) The model parameters of the FUT are tested using the trained ML model according to the conventional method shown in Fig. 1.

A. Exhaustive-FP

The main steps of the X-FP measurement are summarized in Algorithm 3. The inputs of the algorithm are the total grid information (X, Y) and the number of LUT inputs, i. The algorithm returns the exhaustive fingerprint array $\boldsymbol{F}_{\text{X-FP}}$. In X-FP, fingerprint measurements are conducted as shown in lines 2–7. Using the XOR- and XNOR-based ROs as with XP, the measured frequency array of the nth path is formed as $\boldsymbol{F}_n = [f_{n;(1,1)}, f_{n;(2,1)}, \ldots, f_{n;(X,Y)}]$. The fingerprint procedure is iterated to analyze all paths (i.e., 2^{i-1} paths) by the for loop for n. After completing the fingerprinting for all paths, the exhaustive fingerprint array $\boldsymbol{F}_{\text{X-FP}}$ is derived as follows:

$$\boldsymbol{F}_{\text{X-FP}} = [\boldsymbol{F}_1, \boldsymbol{F}_2, \ldots, \boldsymbol{F}_{2^{i-1}}]. \tag{3}$$

The total number of frequencies is $X \times Y \times 2^{i-1}$ in $\boldsymbol{F}_{\text{X-FP}}$, and it is much larger than the number of frequencies for XP and FP.

While the X-FP characterization captures the aging effect on the FUT, a huge amount of feature vectors is yielded, which cannot be appropriately handled by the ML algorithm. For instance, considering the Xilinx Artix-7 FPGA [28], a seven stage RO using a single input LUT as in [12] needs over 3000 ROs to create the FPGA fingerprint shown in Fig. 4.

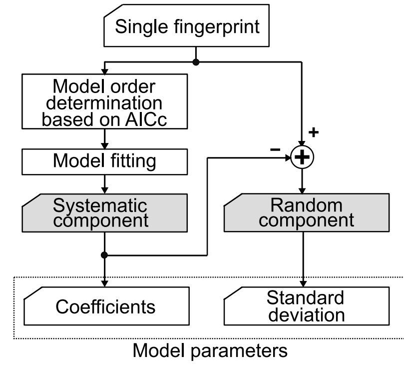


Fig. 6. WID modeling flowchart.

Since the FPGA has 6-inputs LUTs, 32 paths are analyzed per CLB. Hence, the number of frequencies generated is at least 96 000 using the X-FP measurement. As a result, the ML algorithm cannot handle such huge feature vectors appropriately. In our preliminary experiment using Scikit-learn [30], an ML model could not handle the 96 000 feature vectors. According to Occam's Razor principle, a simple model with a few feature vectors is more desirable to achieve better generalization performance in ML algorithms [31]. In the proposed method, the model parameters, which were extracted through WID modeling for each fingerprint path, as seen in [15], were used to overcome the problem of huge feature vectors. This is shown in parts 2) and 4) of Fig. 5.

B. Feature Engineering Based on WID Modeling

Traditionally, manufacturing process variations on semiconductors can be divided into two components: 1) the systematic and 2) the random components [23]. The WID modeling systematically decomposes the process variation into each component from measurement results. In the proposed method, the ML model learns features using the model parameters extracted through WID modeling. Using the proposed method, the dimensionality of the feature vector can be drastically reduced, while preserving the characteristics of the WID process variation of the spatial correlation and the random variation on the die. Thus, $\boldsymbol{F}_{\text{X-FP}}$ becomes applicable to the ML-based classification. Furthermore, the database maintenance also becomes easier. As $\boldsymbol{F}_{\text{X-FP}}$ is composed of 2^{i-1} fingerprints, WID modeling is conducted for each fingerprint.

The WID modeling flowchart for one fingerprinting measurement is shown in Fig. 6. The systematic component is considered as a gradual variation and modeled by the polynomial of the grid location on a die [19], [20]. The random component is modeled as random variables caused by physical phenomena. Considering the geometry of the die, the measured frequency $f_{(x,y)}$ at the location (x, y), (i.e., each frequency in an FPGA) can be expressed as follows:

$$f_{(x,y)} = s_{(x,y)} + r_{(x,y)} \tag{4}$$

where $s_{(x,y)}$ and $r_{(x,y)}$ are the systematic and random components at the location (x, y), respectively. After determining the systematic component $s_{(x,y)}$ and $r_{(x,y)}$ is derived as a residual between the measured frequency and its systematic component. The random component is represented as a normal distribution with zero mean [i.e., as $\mathcal{N}(0, \sigma_{\mathrm{rnd}}^2)$], where σ_{rnd} is the standard deviation of the random component.

However, the determination of the maximum order of the polynomial, which represents the systematic component is not trivial. In the WID modeling, the quality of the fitting is an important criterion that evaluating whether the selected model accurately captures the systematic variation or not. Although the higher-order polynomial gives a well-fitted model with less residual but increases model complexity and consequently it inclines to "over fitting" problem. This type of problem is formulated as "model selection" in the ML field. In order to solve this problem, various information criteria, such as AIC, Bayesian information criterion (BIC) [32], and Watanabe–Akaike information criterion (WAIC) [33], have been proposed. In the proposed method, the most suitable order is derived on the basis of AIC as proposed in [21]. The other information criteria can also be applicable. We utilize an improved version of the AIC, called $\mathrm{AIC_c}$ [34], to obtain a suitable order for the polynomial in the fitting. The $\mathrm{AIC_c}$ is an indicator to balance between the complication of the polynomial order and goodness of fitting. The $\mathrm{AIC_c}$ is expressed as follows:

$$\mathrm{AIC_c} = \log\left(\frac{1}{X \cdot Y}\sum_{x=1,y=1}^{X,Y}\left(s_{(x,y)} - f_{(x,y)}\right)^2\right) + 1 + \frac{2(k+1)}{X \cdot Y - k - 2} \quad (5)$$

where k is the number of the model parameters used in the polynomial. In (5), the first term represents the quality of the fitting which determines the residual, i.e., whether the systematic component is well-fitted or not, while the other terms represent the complexity of the model or the penalty which discourages the overfitting in the higher order of polynomial. The fitting accuracy and the complexity of the polynomial can be balanced using (5). The preferred WID model is considered one with the lowest $\mathrm{AIC_c}$ value discussed in [21]. At this point, it is noted that the order determined from the golden FPGAs is also used for the WID modeling of the FUTs as in step 4) of Fig. 5.

If we find that the third order equation of the systematic component is the best predictable model from the $\mathrm{AIC_c}$ value as an example, then the third-order polynomial of the systematic component is written as follows:

$$s_{(x,y)} = a_0 + a_1x + a_2y + a_3x^2 + a_4xy + a_5y^2 + a_6x^3 + a_7x^2y + a_8xy^2 + a_9y^3 \quad (6)$$

where a_0–a_9 are coefficients. Fitting the frequencies to the polynomial, the model parameters a_0–a_9 can be derived. In the proposed recycled FPGA detection method, the coefficients and σ_{rnd} are utilized for the training and testing in the ML algorithm. Thus, in this example, the length of the feature vector becomes 11 per one fingerprint according to Fig. 6.

The proposed feature vector is capable of approximating the process variation in golden FPGAs simply and accurately. Since the fingerprint largely changes as FPGA is used over time, the feature vector also changes. Thus, the model parameters work well in the ML-based detection.

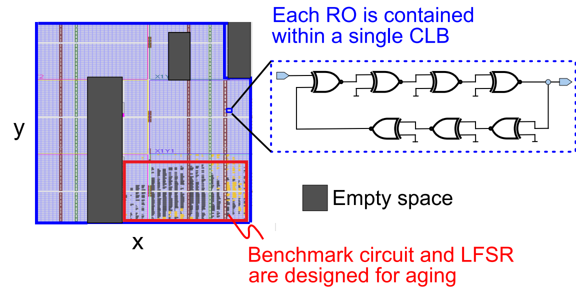


Fig. 7. Floor-plan of Artix-7 with benchmark and LFSR. A single exhaustive-based RO is designed in a CLB. During aging acceleration, the benchmark circuits are placed at the right-bottom corner.

IV. Experimental Results

The effectiveness of the proposed method is determined by conducting measurements using the Xilinx Artix-7 FPGA [28], which is manufactured in 28-nm process technology nodes. Based on measurement results, the proposed recycled FPGA detection method was compared against the conventional method. In this experiment, 50 FPGAs (FPGA-01 to FPGA-50) were used.

A. Measurement Setup

The floor-plan of Artix-7 is shown in Fig. 7. In the floor-plan, most of the area is occupied by CLBs. There is some empty space area, where no logic resource is placed. An on-chip measurement circuit has been implemented using seven-stages (i.e., 7 LUTs) in a CLB. The RO is based on XNOR or XOR logic gates. By keeping the same internal routing, logic resources, and structure for each RO is placed in a CLB so that frequency variation caused by internal routing difference can be minimized. Total 3173 ROs were placed on a geometrical grid of 33×120 (except the empty space of the layout) through hardware macro modeling. Measurements are taken from three different sessions to cover all LUTs and place ROs to FPGAs in order to minimize the routing difference between ROs and MUX. Using hardware macro modeling, initially, ROs are placed over a portion of the LUTs. Then, ROs are moved to cover other remaining portions of the LUTs in the second and third sessions, respectively, to complete the measurement. The purpose was to represent the total layout of the FPGA using the Xilinx CAD tool, Vivado [35]. Aritx-7 FPGA has 6-input LUTs, which create a total of 32 ($= 2^{6-1}$) possible paths for XNOR- and XOR-based RO configuration (path-01 to path-32). In total, to complete the X-FP measurement 32 fingerprint measurements were conducted for all 32 paths. All measurements of the fresh FPGAs are taken at room temperature and the supply voltage of 1.0 V with the help of 100 MHz on-board clock frequency.

The implemented frequency measurement system is shown in Fig. 8. A host PC controls the measurement system through a joint test action group (JTAG) circuit. Each RO is enabled by the selector and the counter circuit measures the frequency

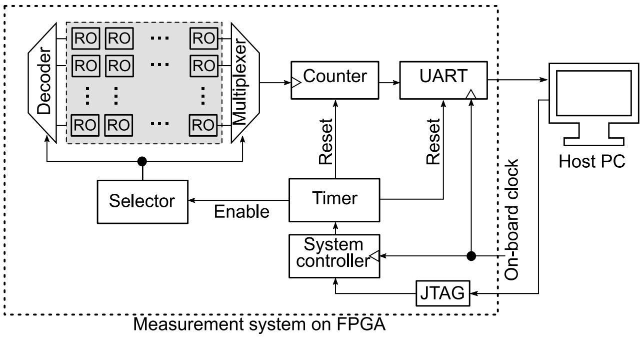


Fig. 8. Block diagram of a frequency measurement system for X-FP.

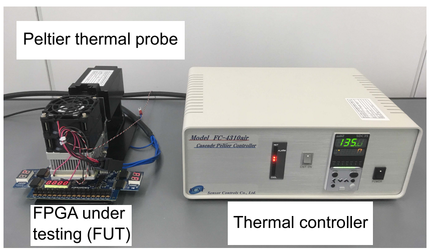


Fig. 9. Experimental setup for accelerating the aging process.

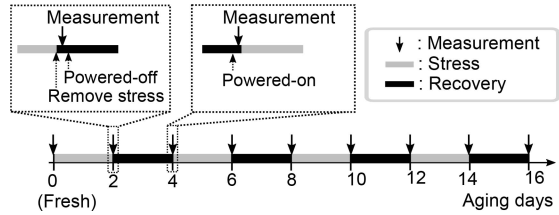


Fig. 10. Aging schedule and measurement timings of the fingerprints of FPGA-01 and FPGA-02. The measurements are conducted at the beginning and the end of the recovery phase.

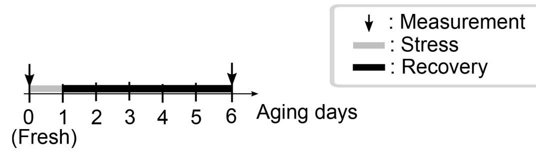


Fig. 11. Aging schedule and measurement timings of the fingerprints of FPGA-03 to FPGA-06.

corresponding to each RO. The selector and counter circuits are controlled by the generated on-chip clock of 100 Hz. The on-chip clock acts as a timer signal and it is made with the help of on-board system clock frequency of 100 MHz. A millisecond pulse is created as a phase-locked loop (PLL) in the system controller to synchronous all generated on-chip clock. Thus, the latency in the on-chip clock can be reduced. At first, each RO is enabled by the selector of the MUX and count the frequency by the counter at a prespecified time period controlled by the timer signal. Then in another signal of the timer, RO is stopped and the measured frequency is transferred to the PC. Similarly, timer is used to reset the counter. This cycle is repeated for taking the measurement from all ROs one by one in order to avoid power or signal interference from other ROs. To collect and store measured frequencies, a universal asynchronous receiver transmitter (UART) module was implemented as an interface between the FPGA and the PC using the Python programming language. The UART is also controlled by the timer. This same measurement system was implemented for all 50 FPGAs.

Among the 50 FPGAs, only 6 FPGAs (FPGA-01 to FPGA-06) were used as recycled FPGAs. The s9234 benchmark circuit from ISCAS'89 benchmark circuit [36] was used, and random workloads were fed to the circuit by a 16-bit linear feedback shift register (LFSR) as a user circuit. These circuits are placed on the right bottom corner of the floor-plan as shown in Fig. 7.

To achieve aging acceleration, a Peltier thermal module (shown in Fig. 9) was used in the experiments. The first two FPGAs (FPGA-01 and FPGA-02) were used both analyzing the aging effect of various aging conditions in the X-FP analysis and also as the recycled FPGAs. The aging schedule used for only two FPGAs is summarized in Fig. 10. The measurements of FPGA-01 and FPGA-02 were carried out by repeating "stress" and "recovery" phases, taking into account the actual conditions. The phases are switched every two days. The measurement cycle continued for 16 days. The fingerprint measurements were conducted only at room temperature at the beginning and the end of the recovery phases. The states of the FPGAs at the beginning and the end of the recovery phases during fingerprint measurement are called the "stress state" and the "recovery state," respectively. During the stress phase, the aging conditions were accelerated by increasing the temperature and switching activities on the s9234 circuit. The FPGA with a maximum rated temperature of 120 °C was heated up to 135 °C using the thermal module. On-board 100 MHz supplied clock was used to achieve dynamic stress from the LFSR circuit. After the stress phase, all stresses were removed and immediately after the FPGA was set back to its normal condition, the fingerprint of the FPGA was measured (stress state). The circuit was then totally powered-off until the next stress phase and then it entered the recovery phase. Before starting the stress phase, the fingerprint of the FPGA was measured again during the recovery phase. Besides, in order to increase further the numbers of recycled FPGAs, the other 4 FPGAs (FPGA-03 to FPGA-06) were used for aging stress (24 h) and measurements were taken after five days recovery phase as shown in Fig. 11.

Regarding the thermal acceleration, we can estimate actual usage time from the artificial stress time using the thermal equipment according to [37]. The thermal acceleration factor F_{T} due to the high temperature can be calculated to estimate the real-time aging of FPGA discussed in [38] as $F_{\mathrm{T}} = e^{\frac{E_a}{k}(1/T_{\mathrm{op}} - 1/T_{\mathrm{stress}})}$. when considering the parameters as activation energy $E_a = 0.5$ eV, Boltzmann's constant $k = 8.62 \times 10^{-5}$ eV/K, nominal operating junction temperature $T_{\mathrm{op}} = 313$ K (40 °C) and stressed temperature $T_{\mathrm{stress}} = 408$ K (135 °C) where the maximum junction temperature is 125 °C [28]. However, as both the stress and recovery phases are involved in our aging schedule, it is hard to calculate the actual usage time accurately. Therefore, it can be roughly estimated to determine the actual usage time. For an example,

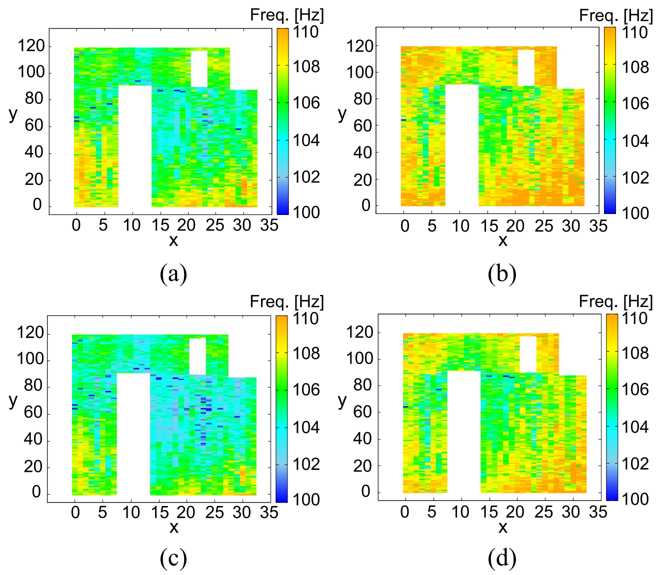


Fig. 12. Fingerprints of path-22 of FPGA-01 and FPGA-02. (a) FPGA-01 (Fresh). (b) FPGA-02 (Fresh). (c) FPGA-01 (16th day). (d) FPGA-02 (16th day).

accelerated aging duration of 24 h can be considered 73 days of actual usage time.

MATLAB software running in the host PC was used to apply the WID modeling to the measured fingerprints. The Python programming language was also used to implement the OC-SVM model using Scikit-learn to enable learning from the features of the fresh fingerprints as well as to make a prediction from the FUT.

B. Results

1) Measurement Results: Fresh and aged fingerprints of path-22 of FPGA-01 and FPGA-02 are shown in Fig. 12. It can be observed that the frequency distributions in the fresh and aged fingerprints have a smooth variation along the coordinates and the fingerprints are different. Thus, they can be used as fingerprints to distinguish each other as proposed in [12]. Another observation is that the distinction is maintained even though the fingerprint is aged.

To observe the aging degradation of the fingerprints, the frequency differences of the 32 paths between the fresh and aged FPGA-01 and FPGA-02 were calculated. Among the 32 paths, path-18 and path-22 of FPGA-01 and path-12 and path-29 of FPGA-02 in the stress state (2nd, 6th, 10th, and 14th days) and recovery state (4th, 8th, 12th, and 16th days) are shown in Figs. 13 and 14, respectively. Path-22 of FPGA-01 and path-12 of FPGA-02 exhibit the largest deviations from the fresh ones, while path-18 of FPGA-01 and path-29 of FPGA-02 exhibit the smallest deviation. From the above figures, it can be easily observed that each of the paths ages differently and these degradations increase as the aging time increases. As shown in Fig. 13(a), (e), (i), and (m), after two days of aging, the aging effect can be clearly observed in the benchmark area in all paths, where the benchmark area is degraded differently. The degradation is larger than other unused areas due to the switching activities of the input in s9234.

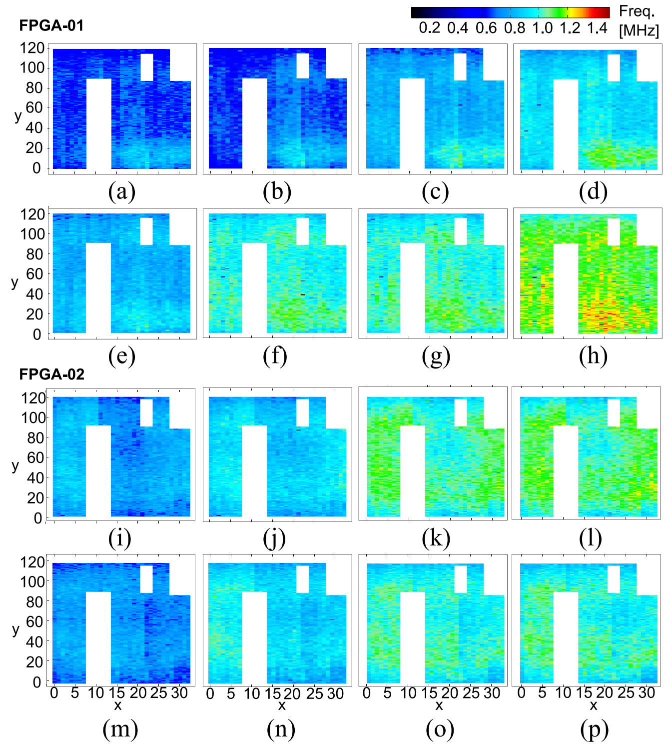


Fig. 13. Path difference between the fresh and aged of FPGA-01 and FPGA-02, respectively, after different aging days at 2nd, 6th, 10th, and 14th day (stress state).

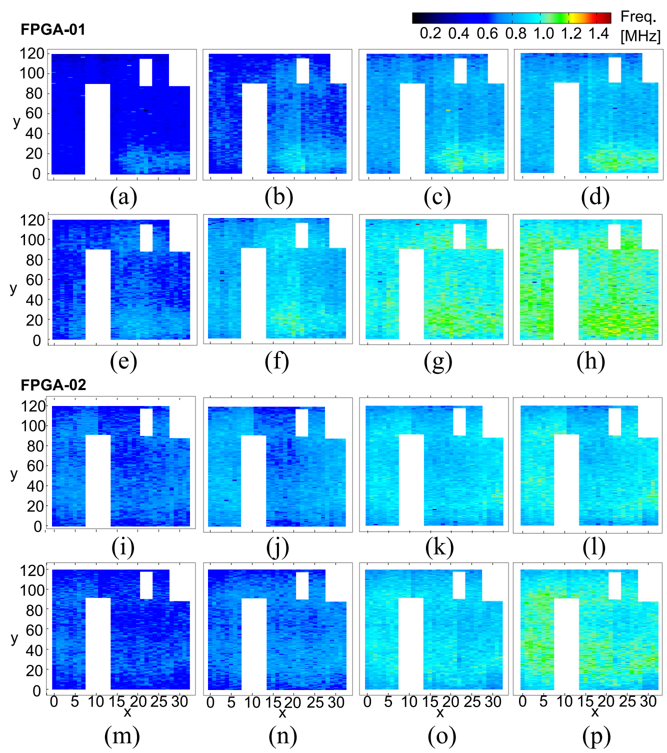


Fig. 14. Path difference between the fresh and aged of FPGA-01 and FPGA-02, respectively, after different aging days at 4th, 8th, 12th, and 16th day (recovery state).

It can be generally assumed that the recycled FPGAs are tested in the recovery state, since they are powered-off when entering the supply chain. Therefore, only the recovery results

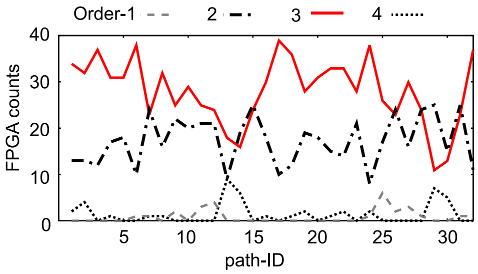


Fig. 15. FPGA counts with the minimum $\mathrm{AIC_c}$ values, when the first to fourth polynomial order is used in the 32 Paths.

shown in Fig. 14 will be discussed. First, it should be noted that the aging effect in the benchmark area can still be observed even though the stress is removed. This is shown in Fig. 14(a)–(d) and (p). On the other hand, no significant difference can be observed in the unused areas. Thus, if the selected CLBs are not appropriately distributed on the FPGA during the XP characterization, the recycled FPGA detection may fail. In addition, the aging trend differs significantly for each device and the unused areas also exhibit less degradation than the benchmark area. This has been reported in [39] and shown in Fig. 14(d), (h), (i), and (p). This result suggests that it may be difficult to successfully detect recycled FPGAs if the appropriate paths are not selected during the FP characterization. Hence, the X-FP characterization, which allows the exhaustive analysis of the logic blocks, is important for achieving an accurate detection of recycled FPGA.

2) WID Modeling: Next, WID modeling is applied to the measurement results in order to extract the model parameters. To determine the order of the polynomial of the systematic component, $\mathrm{AIC_c}$ values were calculated using (5) for each of the 32 paths of the 50 FPGAs used in this experiment. The number of FPGAs with the minimum $\mathrm{AIC_c}$ was counted by varying the order from 1 to 4 for each path. The FPGA counts corresponding to the minimum $\mathrm{AIC_c}$ values for different polynomial orders for each path are shown in Fig. 15. It can be observed that the third-order polynomial model is the best choice to represent the systematic component of the FPGAs in most cases. Although there are some paths for which other orders are more suitable, there is little difference when using the third-order polynomial in most cases. Thus, the third-order polynomial which represents the systematic component, is used to uniformly handle the feature vectors. Through the WID modeling using the third-order polynomial model, the 11 model parameters (i.e., a_0–a_9 of (6) and σ_{rnd}) can be extracted from the fingerprints of all FPGAs.

Using the third-order polynomial, the fingerprints shown in Fig. 12(a) and (c) are decomposed into the systematic and random components, as is shown in Fig. 16. The systematic and random components of the fresh FPGA-01 are shown in Fig. 16(a) and (c), respectively. As a fitting result, the root mean squared errors (RMSEs) of the 32 paths for each of the 50 FPGAs was evaluated. An average of 0.57-MHz RMSEs

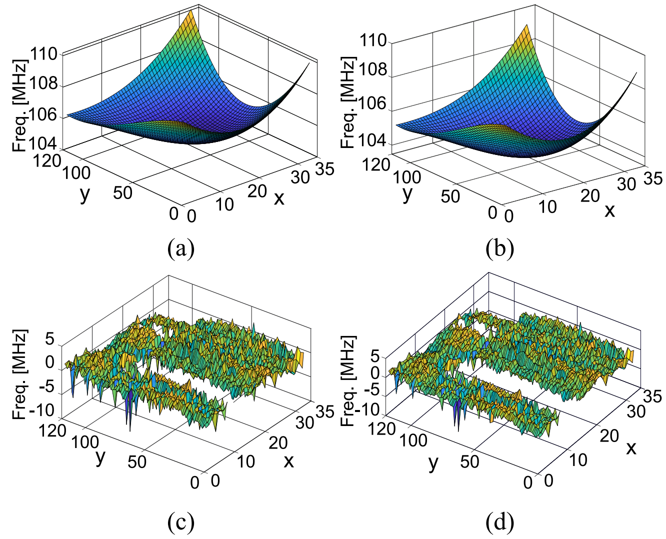


Fig. 16. Decomposition of the WID process variations of path-22 of FPGA-01 when the fresh and the 16th day. The systematic aged component is degraded approximately 1.5 MHz than the fresh one shown in the top left and right corner. (a) Systematic (Fresh). (b) Systematic (Aged). (c) Random (Fresh). (d) Random (Aged).

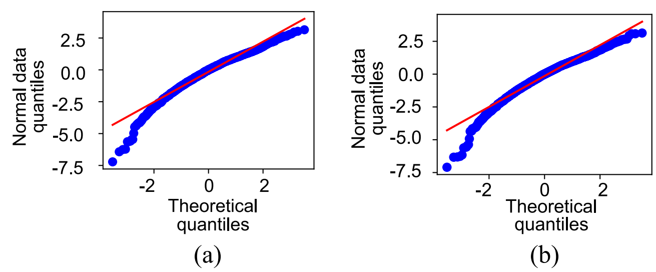
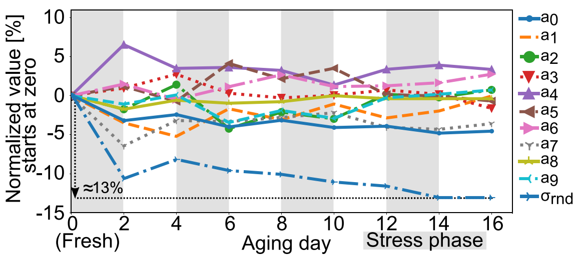


Fig. 17. Q-Q plots of the path-22 residuals of FPGA-01. The residuals follow a normal distribution for the fresh and the 16th day cases, since most of the dots lie on a line with a slope of one. (a) Fresh. (b) Aged.

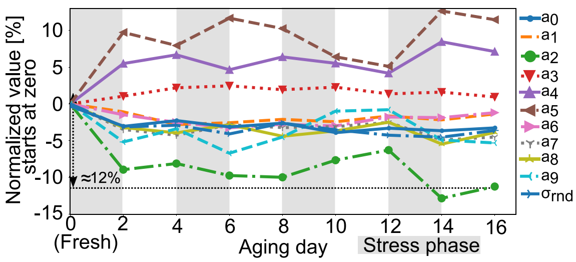
was found for the fresh FPGAs. Here, the RMSE is defined as

$$\mathrm{RMSE} = \sqrt{\frac{1}{32 \cdot X \cdot Y}\sum_{n=1}^{32}\left(\sum_{x=1,y=1}^{X,Y}\left(s_{n;(x,y)} - f_{n;(x,y)}\right)^2\right)} \quad (7)$$

where $s_{n;(x,y)}$ is the systematic component of the frequency of the nth path. Since the fingerprint frequencies of path-22 of FPGA-01 are distributed from 98 to 112 MHz as shown in Fig. 12 and other paths also have similar distributions, the RMSE is very small compared to the process variation distribution. The systematic and random components of the fresh and aged (16th day) conditions are shown in Fig. 16(b) and (d), respectively. It can be observed that the aged systematic component is slightly lower than the fresh one. The average RMSEs of all paths of the aged (16th day) FPGA-01 and FPGA-02 are 0.42 MHz and 0.38 MHz, respectively. Also, the fitting accuracy of the systematic component is improved compared to that of the fresh ones. The normalities of the random components of the fresh and aged FPGAs are evaluated using a quantile–quantile (Q–Q) plot as shown in Fig. 17. It can be observed that the random components are well modeled by a normal distribution for

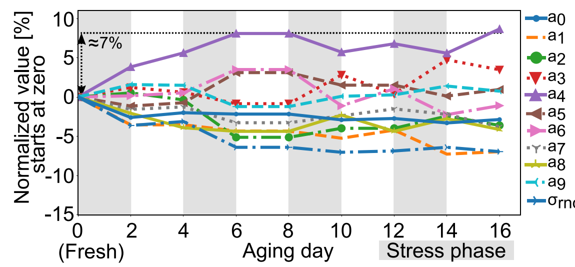


(a)

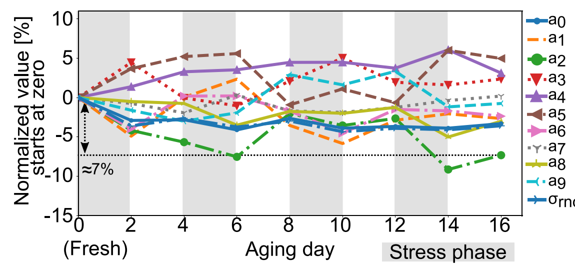


(b)

Fig. 18. 11 model parameter variations of FPGA-01 and FPGA-02 as a function of the aging day (maximum deviation).



(a)



(b)

Fig. 19. 11 model parameter variations of FPGA-01 and FPGA-02 as a function of the aging day (minimum deviation).

the fresh and aged FPGAs. The model parameters extracted through the WID modeling successfully represent the process variation of the FPGAs and still work well for the aged FPGA.

Using WID modeling, 11 parameters were extracted from each path of the X-FP. Each path consists of 3173 frequencies. There are 32 paths in X-FP and each path contains 11 parameters. The total number of parameters used as features in the X-FP is 352 (= 11 × 32), which is subtracted from 101 536 (= 3173 × 32) frequencies. Hence, the result of the dimensionality reduction is 99.6% (= (101536 − 352)/101536). Thus, the proposed feature engineering contributes not only to the database size reduction but also to the prevention of the curse of dimensionality problem in the ML classification.

In general, the training and test data are preprocessed to improve the accuracy of ML. In this experiment, a min-max normalization was applied using the model parameters of the fresh 50 FPGAs as preprocess, i.e., all the fresh parameter values are limited between 0 and 1. The temporal normalized parameter variations of FPGA-01 and FPGA-02 are shown in Figs. 18–20. All lines in the figures are shifted to the start at zero value in order to clearly demonstrate the time dependence variation.

In Fig. 18, path-22 of FPGA-01 and path-12 of FPGA-02 are shown because they exhibit the maximum parameter deviation among the 32 paths. During the stress phase, large variations of some parameters can be observed in both FPGAs. In particularly, σ_{rnd} reaches 13% and 12% of the normalized range in FPGA-01 and FPGA-02, respectively. This is shown in Fig. 18(a) and (b), respectively, at the 16th day, where the maximum deviation is observed compared with other paths of the FPGA-01 and FPGA-02. Similarly, the minimum parameter deviation is shown in Fig. 19, where path-18 of FPGA-01 and path-29 of FPGA-02 are shown. At the 16th day, all parameters in the recovery states (on even number days) deviate at most by 7% from the fresh ones for both FPGA-01and FPGA-02. It is observed that the temporal variations of the model parameters are also different for each path.

The variation of 352 model parameters of FPGA-01 and FPGA-02 at various aging days are shown in Fig. 20. In this figure, the maximum and minimum deviated parameters observed in Figs. 18 and 19 are highlighted. From Fig. 20, it can be observed that the parameters in path-18 of FPGA-01 and path-29 of FPGA-02 have deviated much less than the parameters in the other paths at the 16th day. Notably, the model parameters in the maximum deviated path are as approximately double those in the minimum deviated paths, as is shown in Fig. 20(b). The accurate detection of a recycled FPGA might be difficult when the RO measurement circuit use only path-18 of FPGA-01 or path-29 of FPGA-02. This result greatly supports the proposed method, which uses X-FP characterization and model parameters aiming at maintaining or improving the accuracy of the ML-based classification.

3) Recycled FPGA Detection: Finally, the recycled FPGA detection accuracy is evaluated using the extracted model parameters. In this experiment, only the X-FP and FP-based recycled FPGA detection methods are compared as both of these methods are considered all LUTs. On the other hand, without considering all the LUTs as XP, it may hard to observe the aged part in the FPGA, if the target CLBs are appropriately selected, as shown in Fig. 13(a). Therefore, the proposed method is not compared with the XP-based recycled FPGA detection.

Regarding the FP-based recycled FPGA detection method, a single path RO measurement for each of the 32 paths was conducted. Measurements of the training and tested FPGAs were conducted separately to simulate a real situation, where the FPGAs are measured upon production and then are measured after been used to identify whether they are fresh or recycled. Initially, an OC-SVM model from the fresh 50 FPGAs

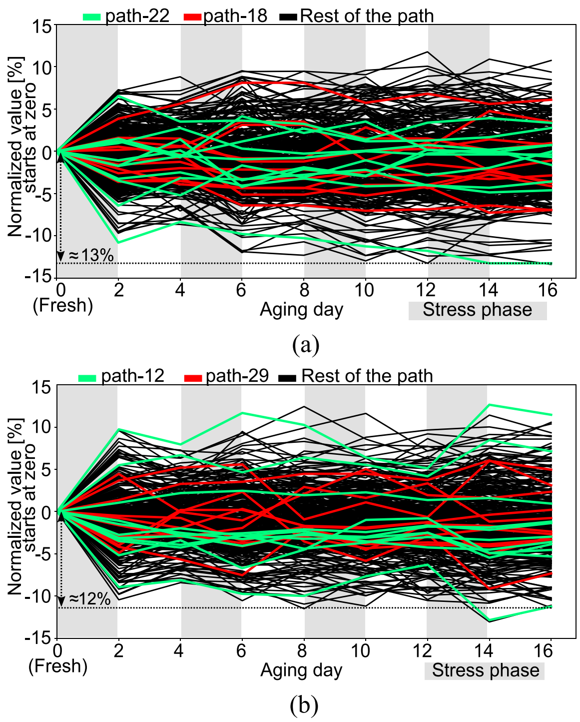


Fig. 20. 352 model parameter variations for all paths as a function of the aging day. The model parameters of the maximum and minimum deviate paths in each FPGA, i.e., path-18 and path-22 of FPGA-01 and path-29 and path-12 of FPGA-02, are highlighted. (a) FPGA-01. (b) FPGA-02.

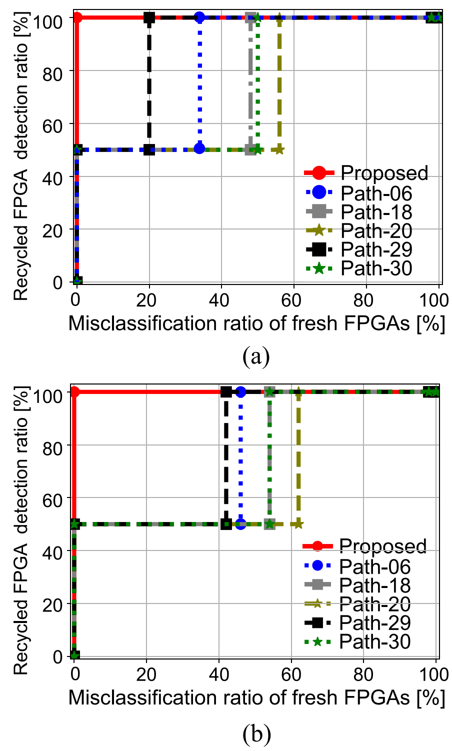


Fig. 21. ROC curves of the proposed recycled FPGA detection method compared with the FP-based method at 4th and 16th days. Using the proposed method, the two aged FPGAs are detected at 100% correct classification ratio of the fresh FPGAs. ROC of path-06, path-18, path-20, path-29, and path-30 shows the performance degradation. (a) 16th day. (b) 4th day.

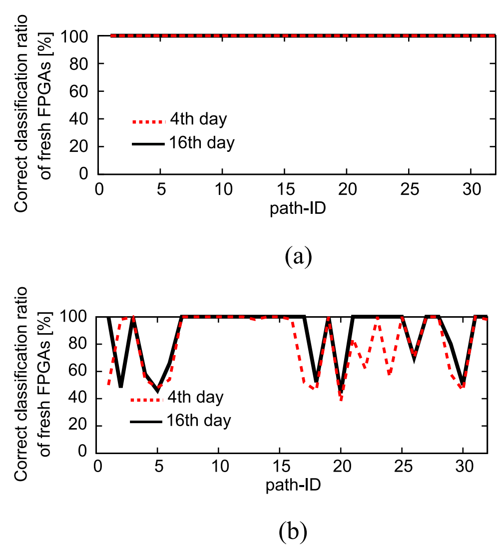


Fig. 22. Correct classification ratios for the fresh FPGAs, when the aged (a) FPGA-01 and (b) FPGA-02 are detected accurately using the FP-based recycled FPGA detection method. It is noted that the proposed method achieves a 100% correct classification ratio.

was trained. This model contains 352 features for each FPGA in total. Then, all fresh FPGAs and the two aged FPGAs (FPGA-01 and FPGA-02) were tested at the 4th and 16th days. It can be expected that the detection in the 16th day is easier than the detection in the 4th day. The hyper-parameters of the OC-SVM model in the training data set were selected empirically, as no significant accuracy improvement was observed by further increasing in the parameter values. The radial basis function (RBF) is used as a kernel function with $\gamma = 0.015$ and parameters $\nu = 0.01$ which defines both the upper bound on the fraction of the outliers and lower bound of the fraction of the support vectors [30].

The receiver operating characteristic (ROC) curve is presented in Fig. 21 to show the classification result of the proposed method compared with the FP-based method in the 4th and 16th days for the two aged FPGAs (FPGA-01 to FPGA-02). Among 32 paths in the FP-based method, found the worst scenarios ROC results using the path-06, path-18, path-20, path-29, and path-30 as described later. Therefore, the proposed method is compared with only using those paths. In the figure, "Recycled FPGA detection ratio" means a true positive rate, and "Misclassification ratio of fresh FPGAs" means a false positive rate.

Using the proposed method, the ROC curve indicates a very good performance for the detection of both aged FPGAs, where the proposed method exhibits no misclassification for the fresh FPGAs. In contrast, some misclassifications can be observed using the FP-based method. In Fig. 21, ROC curves of the FP-based method using path-06, path-18, path-20, path-29, and path-30 indicate a degraded performance. The correct classification ratios of the fresh FPGAs for the FP-based method using the 32 paths individually, when the two aged FPGAs are identified as aged, is shown in Fig. 22. In the figure, "Correct classification ratio of fresh FPGAs" means a true negative rate when the aged FPGA is detected perfectly. As also shown in Fig. 22(a), the FP-based method

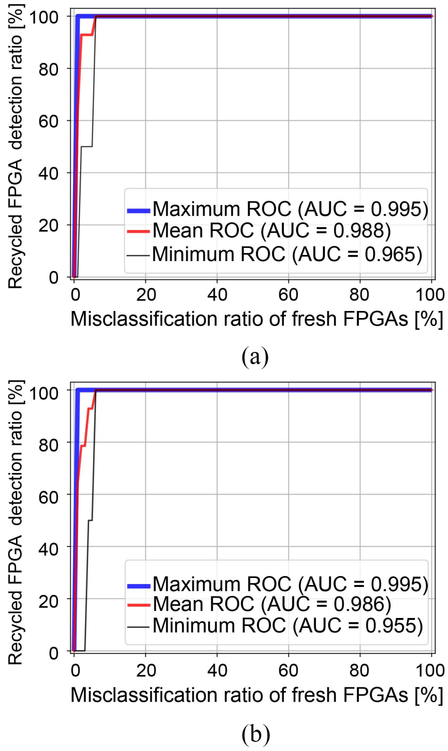


Fig. 23. ROC curves at seven different training data sets show the detection rate of two aged FPGAs (FPGA-01 to FPGA-02) using our proposed method.

accurately classifies the fresh FPGAs and the aged FPGA as does the proposed X-FP-based method for FPGA-01. On the other hand, the aged FPGA-02 is difficult to be detected as is shown in Fig. 22(b). This is because the temporal variations in of FPGA-02 are more largely distributed than those in FPGA-01 as shown in Fig. 20.

Furthermore, although we had considered all the fresh FPGAs in the OC-SVM model training in the above experiments, a more thorough analysis of the training process of the OC-SVM model was conducted to evaluate the performance of the prediction model by using different training sets. In this evaluation, we first randomly divide the 50 fresh FPGAs into seven groups with 8, 7, 7, 7, 7, 7, and 7 fresh FPGAs. Then, six groups selected from the seven groups were used for the OC-SVM model training. This process was repeated seven times, where one group was unused exactly once. Thus, the fresh samples of FPGA-01 to FPGA-06 were randomly contained in some of the training sets. The seven trained OC-SVM models obtained from the training were tested using the 50 fresh FPGAs and aged FPGAs (FPGA-01 and FPGA-02). Fig. 23 shows the ROC curve of detecting the two aged FPGAs on the 16th and 4th day. The red curves show the mean ROC curves calculated through interpolating 100 points over the mean value of the seven different ROC results. The area under the curves (AUCs) of the mean ROC at the 16th and 4th days aging detection are 0.988 and 0.986, respectively. Similarly, the ROC curves having maximum and minimum AUCs of the seven ROC curves are shown by the blue and black lines, respectively. It can be seen that the high accuracy is still observed for the detection of the two aged FPGAs when using different training sets.

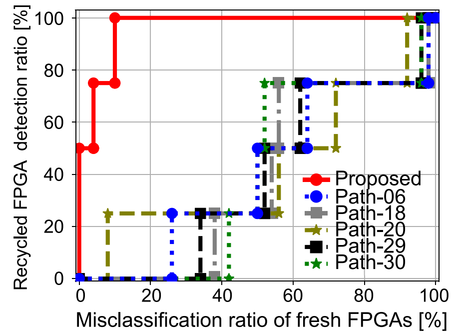


Fig. 24. ROC curves for the four aged FPGAs (FPGA-03 to FPGA-06) which were aged by 24 h. Our proposed method achieves better performance than the FP-based method. The best path of the conventional method is path-20, whose AUC value is 0.430.

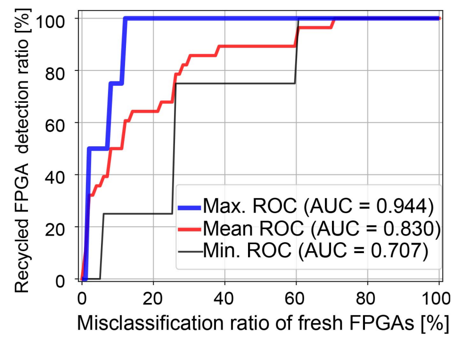


Fig. 25. ROC curves at seven different training data sets show the detection rate of 4 aged FPGAs (FPGA-03 to FPGA-06) using our proposed method.

To observe the more efficacy of the proposed method at the lower aging day, we carried out an additional experiment using more aged FPGAs. Fig. 24 shows the ROC curves of the detection rate of the 4 aged FPGAs (FPGA-03 to FPGA-06) which were aged 24 h as shown in Fig. 11. All the 50 fresh FPGAs were used as the training data as with the experiment of Fig. 21. Using the proposed method, the two aged FPGAs are detected with 100% classification of fresh FPGAs and other two aged FPGAs are detected with misclassification of one and four fresh FPGAs, respectively. On the other hand, the ROC curves of the FP-based method using path-06, path-18, path-20, path-29, and path-30 shows the degraded performance compared to X-FP shown in Fig. 24.

The mean ROC curves of detecting the four aged FPGAs which were aged 24 h using the proposed method are shown in Fig. 25. In this evaluation, we used the same seven trained OC-SVM models to ones used for drawing Fig. 23. The ROC curves are plotted by testing the fresh 50 FPGAs and the four aged FPGAs. The AUC score of the mean ROC is 0.830. The maximum AUC score is found 0.944 indicating that even higher detection accuracy is achieved using only 42 training samples. On the other hand, although minimum ROC curves show a moderate AUC score of 0.707, the performance is still higher than the maximum AUC score of 0.430 (path-20) of the conventional FP-based method shown in Fig. 24, even though the number of training samples used in the OC-SVM is lesser.

From the above results, it can be concluded that the proposed approach, which includes X-FP and the proposed feature vector engineering, is capable of achieving better classification, whereas the conventional method exhibits some misclassification.

V. Conclusion

In this work, we proposed an accurate recycled FPGA detection method. The proposed method employs an RO-based X-FP path analysis which allows exhaustively to capture the aging effect on all paths in FPGA although it generates a large number of feature vectors. Since the proposed method achieves a drastic reduction on the number of feature vectors by the WID-modeling-based feature engineering, the X-FP analysis enables the efficient and effective detection of recycled FPGAs. Experimental results using 50 commercial FPGAs confirmed that the X-FP analysis can capture the degradation effects, which cannot be detected by conventional methods. It also shows the effectiveness of the model parameters extracted by the proposed WID modeling in the detection of recycled FPGAs. The model parameters for some of the paths of aged fingerprints exhibited more than 10% differences on different aging days. The OC-SVM was capable of successfully categorizing aged FPGAs with higher detection accuracy with over 0.96 AUC score, while achieving a 99.6% database size reduction for each FPGA.

The evaluation of the X-FP analysis is still limited since its evaluation is aimed at recycled FPGA detection in this work. We intend to investigate a further evaluation of the X-FP analysis as an aging analysis method under various aging scenarios.

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